Pippi de Bree

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Education

New York University, Steinhardt **Expected May 2027**
Master of Science in Applied Statistics for Social Science Research
GPA: 4.0

BrainStation Data Science Bootcamp, London, UK **November 2022**
Admitted with a full scholarship to twelve week bootcamp.

Boston University, College of Arts and Sciences **January 2020**
Bachelor of Arts in Economics and Mathematics, Minor in Computer Science
Dean's List Spring 2018, Fall 2018 and Spring 2019

Peer Reviewed Publications

de Bree, P., Mesalles, V., Morelli, A., Downey, D., Wang, X., Chouthai, M., Lepore, N., Harel, D. Don't worry, I asked ChatGPT: Investigating use of Generative artificial intelligence among sexual and gender minorities and cisgender heterosexual individuals. (Under Review) *Journal of Homosexuality*.

Research Experience

Research Assistant **Summer 2026 - Present**
Supervisor: Dr. Klint Kanopka

Sparse Variational Autoencoders for Item Response-Like Modeling

- Investigated adapting a deep generative model with a variational autoencoder and sparse decoding (through spike and slab priors) to item response theory interpretations.
- Wrote three adaptations of the model in PyTorch and adapted original model code for transparent training. Tuned priors and training paradigm to fit model assumptions.
- Wrote about the relationship between this as a potential method and traditional models used in Item Response theory.

NYU QUEER Data Lab, Research Assistant
Supervisor: Dr. Daphna Harel

- Led statistical analyses and evaluation of methods to derive perceptions from an AI use and sentiment survey, wrote the code to do so in R.
- Conducted a statistical literature review of modeling options and communicated the implications to the lab, leading the selection of the final modeling strategy.
- Fit factor analyses and a multi-level model to compare estimated marginal mean uses and perceptions of AI across sexual and gender minority+ groups.

- Collaboratively wrote the statistical methods, results, and discussion sections of the paper.

Research Assistant **Fall 2025 - Present**
Supervisor: Dr. Marc Scott

What Counts? The Use of History in Sequence Analysis and Prediction

- Set up cross validation paradigms to evaluate a novel method, categorical functional data analysis, against traditional feature engineering methods. Fit both linear and random forest models.
- Evaluated school-to-work transition data and built the framework for a simulation study.
- Performed sensitivity analyses on all methods to ensure a "fair" competition. For example, tuned parameters for soft clustering and build test set assignment function.
- Developed visualizations to compare representations of sequences across time, particularly for time-varying methods.
- Wrote the Methods section and several appendices regarding sensitivity analyses for the paper, and collaborated on writing the remainder of the paper.

The Branching Tree of the Life Course: Incorporating History in Sequences Analysis

- Fit hard and soft clustering, with sequence focused distance metrics, on school-to-work transition data. Fit on multiple subsequences of data, with a fixed transition point, for comparison against a novel method - multi-phase model based clustering with transition points.
- Developed visualizations of state probabilities, across time, for both clustering methods, for comparison against visualizations from model based clustering methods.

Course Projects

Can LLMs Compare to Human Content Moderators? **Fall 2025**
Messy Data and Machine Learning | NYU Course Project for APSTA-GE 2047

- Performed the technical and statistical work of a group project comparing LLM against crowd-sourced ratings of content moderation on X's Community Notes.
- Created a subset of English content moderation notes, and corresponding ratings, with conclusive outcomes (i.e. marked as valid moderation or removed from the platform).
- Implemented keyword assisted topic modeling to reduce data to two controversial topics.
- Developed a prompting pipeline, in R, to generate synthetic content moderation ratings from the LLM. Iterated over prompts to ensure a strict form of output data, to allow for modeling.
- Fit random forest and logistic regression models on human-ratings training data to benchmark the relationship between ratings survey variables and outcomes. Compared performance of test set of LLM survey responses against corresponding withheld human-ratings set.

- Explored limitations of an overfit model due to large differences found in LLM survey responses.

Exploring Recommendation System Suitability by User History Depth **Fall 2022**
Capstone | BrainStation
- Built four recommendation systems in Python to compare users with varying levels of listening history. Recommenders included a popularity based, content based and two collaborative filtering systems based on latent feature processing.
- Presented findings in an interactive presentation with real-time recommendations.

Professional Experience

Quantitative Student Specialist | NYU Data Services **May 2026 - Present**
- Provide cross-disciplinary quantitative research consultations to members of the NYU community, focusing on statistical software support (R, Stata) as well as data cleaning and manipulation.
- Lead R coding tutorial "Data Wrangling in R". Adapted content to modern R practices.
- Collaborate with subject area specialists and data librarians to support research data lifecycle.

Machine Learning Engineer Intern | Durable **Spring-Summer 2023**
Louisville, CO
- Built a Retrieval Augmented Generation model by customizing code of pre-trained Hugging Face encoders and generators.
- Developed pre-training for the retrieval system to ensure encodings took on bespoke language form necessary for integration with code synthesis paradigm. Created bespoke evaluation metrics for pre-training based on the retriever's ranking.
- Designed a training pipeline, using PyTorch Lightning and ClearML reporting.
- Collaborated closely with the full start-up team, of 6, to integrate the model into the product.

Teaching

Teaching Assistantships
Messy Data and Machine Learning | NYU APSTA-GE 2047 Fall 2026
- Led and developed recitation sections for 34 graduate students.
- Set up batch homework autograding and

Cracking the Code |NYU APSTA-UE 0021 Spring 2026
- Developed and led an Introduction to R programming lecture for non-technical undergraduate students.
- Led a lecture and lab sessions teaching students to use the programming language R, in RStudio.

Coursework

Graduate
Frequentist Inference
Bayesian Inference
Causal Inference
Multi-Level Modeling
Supervised and Unsupervised Machine Learning
Messy Data and Machine Learning
Generalized Linear Models
Data Science for Social Impact

Undergraduate
Linear Models
Probability
Linear Algebra
Differential Equations
Combinatronic Structures
Behavioral Economics
Game Theory
Empirical Econometric Analysis I&II

Skills & Technologies

Advanced R (base, tidyverse, ggplot)
Advanced Python (Pandas, Numpy, Scikit-learn)
PyTorch, PyTorch Lightning, SQL, Git, Stata